Ahmad Kassar

Software engineer: full-stack development, AI integration, Linux and cloud infrastructure

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Profile

I build software that companies run every day: a RAG assistant that answers engineers from their own technical documentation at Hyperlean, an ordering platform with its own database rules and AI assistant for a pizzeria in Ancona, and services I deploy myself with Docker on Linux and AWS. BSc in Computer Science from the University of Camerino; now in the MSc programme.

Experience

2026 – now

Software developer, client projects
Pizzeria Buon Gusto, Prontix, Stepvia International Education

- Designed and built the ordering platform and back-office for Pizzeria Buon Gusto: Supabase Postgres with 14 tables, row-level security on every table, server-side pricing, an order state machine enforced in the database, stock, delivery slots and zones.
- Added an OpenAI assistant that answers only from live menu, stock and order data, with model fallback and abuse limits; Web Push and email notifications on every order change; live chat handing customers over to staff.
- Built the Prontix site and client area (Google and email sign-in, document upload, credit-metered questions over customers' documents) and the Stepvia site in English, French and Arabic.

Oct 2025 – Mar 2026

AI chatbot and software developer
Hyperlean S.r.l.

- Designed and built a RAG virtual assistant for an industrial cost-estimation and quoting application: FastAPI backend, Qdrant vector database and OpenAI models, integrated into the existing VB.NET desktop client.
- Built the ingestion pipeline for PDF, Excel and text documentation (extraction, normalisation, chunking, embeddings), fixing noisy PDF and spreadsheet extraction that was degrading retrieval.
- Deployed on AWS EC2 with Docker and layered security (security groups, API keys, UFW); added per-user rate limits and daily token quotas in Redis, model fallback, output caps and a health endpoint.
- Added a multilingual technical glossary applied after generation, separate production and development knowledge bases, and evaluated the assistant with guided conversational tests. The project became my BSc thesis.

Sep 2024 – Feb 2025

Software developer
Food & More di Mahfouz Oussama S.A.S.

- Worked on an order-management application: requirements, core features, functional testing and debugging.
- Built web interfaces in HTML, CSS and JavaScript, integrated a relational database and managed code with Git.

Projects

- **RAG assistant for industrial cost estimation**: A virtual assistant built at Hyperlean that answers questions about an industrial cost-estimation and quoting application using only its own documentation, deployed on AWS with cost and abuse controls. Also my BSc thesis.
- **Pizzeria Buon Gusto**: Online ordering, staff back-office and an AI assistant for a wood-fired pizzeria in Aspio, Ancona, running on its own domain.
- **AI Locale**: A private multi-agent AI assistant on a laptop GPU: nine agents plan, work in parallel and review each other, with real tools that ask before anything risky.
- **Slotbook**: Appointment booking for small service businesses: a REST API on PostgreSQL with roles, rotating sessions and a web client, where double booking is impossible even under concurrent requests.
- **kassar-house-server**: Tooling for a self-hosted home server: deduplicated, verifiable backups with retention and safe restores, health checks that notify only when something changes, and a LAN-only service stack behind one HTTPS proxy.
- **Hostwatch**: A self-hosted monitor for Linux servers: an agent reads the kernel's own numbers, a server stores history and evaluates alert rules, and a dashboard updates live over Server-Sent Events.
- **Prontix**: Website and client area for a software, AI and automation service for small Italian businesses, with sign-in, document upload and credit-metered questions over a client's own documents.
- **Shipyard**: The path to production for Slotbook and Hostwatch on one AWS EC2 host: Docker Compose behind Caddy with automatic HTTPS, verified backups, health-checked deploys with automatic rollback, and a CI/CD pipeline with scanning and a manual approval gate.
- **JTime**: A JavaFX desktop app for planning projects and tracking how long tasks really take, with weekly and monthly time reports and CSV export. A university project I later hardened with tests, CI and data-safety fixes.
- **Stepvia International Education**: A multilingual site for an agency that helps students from Lebanon apply to Italian universities, from choosing a course to arriving with a visa.

Skills

Languages: Python, TypeScript / JavaScript, SQL and PL/pgSQL, VB.NET, Java, C++
AI and data: Retrieval-augmented generation, OpenAI API, embeddings, Qdrant vector database, Document ingestion (PDF, Excel), Cost and abuse controls, Local LLMs and multi-agent tool calling
Backend: FastAPI, PostgreSQL, Supabase: Auth, RLS, Edge Functions, REST APIs and OpenAPI, Redis, Real-time: SSE, Web Push
Linux, cloud and delivery: AWS EC2, security groups, Docker and Docker Compose, Linux hardening (UFW, SSH, updates), CI/CD with GitHub Actions, Netlify

Education

- MSc in Computer Science (LM-18), University of Camerino (Jul 2026 – now)
- BSc in Computer Science (L-31), University of Camerino (2023 – 2026). Thesis: "Design and experimental evaluation of a RAG virtual assistant supporting an industrial quoting tool". Supervisor Prof. Lorenzo Rossi.
- Java Programming and Software Engineering Fundamentals, Duke University on Coursera
- Beginning C++ Programming – From Beginner to Beyond, Online course

Languages

Arabic (native), Italian (C1, professional), English (B2, professional), French (B1, working knowledge)